

20. Implement Future Sales Prediction using a suitable machine learning algorithm

Aim

To implement **Future Sales Prediction** using a suitable machine learning algorithm.

Software Required

- Python 3.x
- NumPy
- Pandas
- Scikit-learn
- Matplotlib

Procedure

1. Create a historical sales dataset.
2. Use time/month as the input feature.
3. Use sales quantity as the target.
4. Train a Linear Regression model.
5. Predict future sales.
6. Display the predicted sales values.
7. Evaluate the model.

Code

```
import numpy as np

from sklearn.linear_model import LinearRegression

months = np.array([
    1, 2, 3, 4, 5, 6,
    7, 8, 9, 10, 11, 12
]).reshape(-1, 1)

sales = np.array([
    100, 110, 120, 130,
    140, 150, 160, 170,
    180, 190, 200, 210
])
```

```
model = LinearRegression()
model.fit(months, sales)
future_months = np.array([
    13, 14, 15, 16, 17, 18
]).reshape(-1, 1)
future_sales = model.predict(future_months)
print("Future Sales Prediction:")
for month, sale in zip(
    future_months.flatten(),
    future_sales
):
    print("Month", month,
          ":", round(sale, 2))
```

Output:

```
Future Sales Prediction:
Month 13 : 220.0
Month 14 : 230.0
Month 15 : 240.0
Month 16 : 250.0
Month 17 : 260.0
Month 18 : 270.0
```

Result

Thus, the **Future Sales Prediction model** was successfully implemented using Linear Regression.